

```
from google.colab import files
uploaded = files.upload()
```



Choose Files advertising.csv

advertising.csv(text/csv) - 4062 bytes, last modified: 9/29/2025 - 100% done
Saving advertising.csv to advertising.csv

```
#Import Libraries
import pandas as pd
import numpy as np
from sklearn.model_selection import train_test_split
from sklearn.linear_model import LinearRegression, SGDRegressor
from sklearn.metrics import mean_squared_error
from sklearn.preprocessing import StandardScaler
```



```
df = pd.read_csv("advertising.csv")
df
```



	TV	Radio	Newspaper	Sales
0	230.1	37.8	69.2	22.1
1	44.5	39.3	45.1	10.4
2	17.2	45.9	69.3	12.0
3	151.5	41.3	58.5	16.5
4	180.8	10.8	58.4	17.9
...	...	...	...	...
195	38.2	3.7	13.8	7.6
196	94.2	4.9	8.1	14.0
197	177.0	9.3	6.4	14.8
198	283.6	42.0	66.2	25.5
199	232.1	8.6	8.7	18.4



200 rows × 4 columns

```
#prepare data
X = df.drop('Sales', axis=1)
y = df['Sales']
#Split Data 25% for test random_state=42 define random state = 42
X_train, X_test, y_train, y_test = train_test_split(X, y, test_size=0.25)
```

```
#Linear Regression
linear_model = LinearRegression()
linear_model.fit(X_train, y_train)
```



LinearRegression ⓘ ?

LinearRegression()



```
y_pred_linear = linear_model.predict(X_test)
mse_linear = mean_squared_error(y_test, y_pred_linear)
print(y_pred_linear)
print()
print(mse_linear)
```



```
[ 9.43622871  5.45967971 21.0543385  23.71128267 16.56087761 12.19331518
16.8480005  15.16802964 10.86466061 10.80351067  6.93431481 17.94323456
21.23024131 10.61803493  8.59930729 14.21271799 19.35489625 21.34632431
20.54926022 11.1311576  13.7713316  17.22804196 17.16403104 12.12705351
15.86825597 21.75929545 10.54184793 10.35003278 18.04467935 21.67219843
15.31493643 20.74962454 15.46221575 10.18194351 12.55141071 18.69106495
18.03719676  5.368617   7.83427402  9.4639358   8.90687519 10.60367485
19.20121249 15.1436897  9.73818829 10.96228748 13.05097207  6.07313014
22.62720454 21.26023889]
```

2.7476902717394953



```
#SGD Regressor
scaler = StandardScaler()
X_train_scaled = scaler.fit_transform(X_train)
X_test_scaled = scaler.transform(X_test)
sgd_model = SGDRegressor(random_state=42, max_iter=1000)
sgd_model.fit(X_train_scaled, y_train)
```



SGDRegressor ⓘ ?

SGDRegressor(random_state=42)

```
y_pred_sgd = sgd_model.predict(X_test_scaled)
mse_sgd = mean_squared_error(y_test, y_pred_sgd)
print(y_pred_sgd)
print()
print(mse_sgd)
```



```
[ 9.38925536  5.47711196 21.02836314 23.61950201 16.53591011 12.21174281
16.82349615 15.13102792 10.85177656 10.74135389  6.93027832 17.99572463
21.16464209 10.62395717  8.58891904 14.17811171 19.31704944 21.29261137
20.54520088 11.09077133 13.75925886 17.21667374 17.17708101 12.16640768
15.82686582 21.69624283 10.51550322 10.33727428 18.01805439 21.57380964
15.31608825 20.69820721 15.4105822  10.18432449 12.53676185 18.65806321
17.9545764   5.36007937  7.84203188  9.42153055  8.91020894 10.59578147
19.14095554 15.12840256  9.71987616 10.89288018 13.05534066  6.05405836
22.61141134 21.20257766]
```

2.768842543094607

```
print("Summary Table")
print("=" * 45)
print(f"{'Model':<20} {'Mean Squared Error (MSE)'}")
print("-" * 45)
print(f"{'LinearRegression':<20} {mse_linear:.6f}")
print(f"{'SGDRegressor':<20} {mse_sgd:.6f}")
```



Summary Table

```
=====
Model                               Mean Squared Error (MSE)
-----
LinearRegression                     2.747690
SGDRegressor                         2.768843
```



```
print("Comparison")
if mse_linear < mse_sgd:
    print("LinearRegression on has a lower MSE (better)")
else:
    print("SGDRegressor on has a lower MSE (better)")

print(f"different MSE = {abs(mse_linear - mse_sgd):.6f}")
```



```
Comparison
LinearRegression on has a lower MSE (better)
different MSE = 0.021152
```